

MSTAR PRACTICE WORKSHEET**Lesson 5-10: Volume of Rectangular Prisms**

Grade 6 Mathematics · Standard 6.GR.2

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 **SELECTED RESPONSE — CORRECT: C**What is the volume of a rectangular prism with $l = 4$ in, $w = 3$ in, $h = 5$ in?

- ☐ A 12 in³
- ☐ B 24 in³
- ☒ C 60 in³
- ☐ D 60 in²

$$V = l \times w \times h = 4 \times 3 \times 5 = 60 \text{ cubic inches.}$$

Question 2 **SELECTED RESPONSE — CORRECT: A**A box has a volume of 36 cm³. Its length is 6 cm and width is 3 cm. What is the height?

- ☒ A 2 cm
- ☐ B 3 cm
- ☐ C 6 cm
- ☐ D 12 cm

$$V = l \times w \times h \rightarrow 36 = 6 \times 3 \times h \rightarrow 36 = 18h \rightarrow h = 2 \text{ cm.}$$

Question 3 **SELECTED RESPONSE — CORRECT: A**

How is volume different from area?

- ☒ **A** Volume measures 3D space (cubic units); area measures 2D surface (square units)
- ☐ **B** Volume uses square units; area uses cubic units
- ☐ **C** Volume only applies to cubes; area applies to all shapes
- ☐ **D** There is no difference

Volume measures the space inside a 3D shape (cubic units like in^3). Area measures a flat 2D surface (square units like in^2).

Question 4 **SELECTED RESPONSE — CORRECT: A**

Two boxes: Box A is $10 \times 5 \times 4$ inches. Box B is $8 \times 6 \times 5$ inches. Which has more volume and by how much?

- ☒ **A** Box B by 40 in^3
- ☐ **B** Box A by 40 in^3
- ☐ **C** Box B by 60 in^3
- ☐ **D** They're equal

Box A: $10 \times 5 \times 4 = 200 \text{ in}^3$. Box B: $8 \times 6 \times 5 = 240 \text{ in}^3$. Box B is bigger by 40 in^3 .

Question 5 **SELECTED RESPONSE — CORRECT: B**

What is the volume of a prism with edge lengths $\frac{1}{2}$ m, 3 m, and 4 m?

- ☐ **A** 3 m^3
- ☒ **B** 6 m^3
- ☐ **C** 7.5 m^3
- ☐ **D** 12 m^3

$V = \frac{1}{2} \times 3 \times 4 = 6 \text{ m}^3$.

Question 6

SELECTED RESPONSE — CORRECT: D

A cube has an edge length of 5 cm. What is its volume?

- ☐ A 15 cm³
- ☐ B 25 cm³
- ☐ C 75 cm³
- ☒ D 125 cm³

A cube's volume is $\text{edge}^3 = 5 \times 5 \times 5 = 125 \text{ cm}^3$.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7

TWO-PART QUESTION

Part A — correct answer: C

A container measures $1\frac{1}{2}$ feet long, 2 feet wide, and 3 feet tall. What is its volume?

- ☐ A 6 cubic feet
- ☐ B $6\frac{1}{2}$ cubic feet
- ☒ C 9 cubic feet
- ☐ D 18 cubic feet

$1\frac{1}{2} = 3/2$. $V = 3/2 \times 2 \times 3 = 3 \times 3 = 9$ cubic feet.

- **A:** The $1\frac{1}{2}$ was rounded down to 1. Use $3/2$ for the length: $3/2$ times 2 times 3.
- **B:** The three measurements were added. Volume multiplies length, width and height.
- **D:** That is twice the correct volume, as if the length were 3 feet rather than $1\frac{1}{2}$.

Part B — correct answer: C

Why is a mixed number converted to an improper fraction before multiplying edge lengths?

- ☐ A Because improper fractions are always larger.
- ☐ B Because volume cannot be a fraction.
- ☒ C So the whole-number part and the fraction part are both multiplied, not just one of them.
- ☐ D Because mixed numbers cannot be multiplied at all.

Multiplying only the whole-number part drops the fractional part of the edge. Writing $1\frac{1}{2}$ as $3/2$ keeps the entire length in the product.

Question 8

WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Dropping the Fraction

Omar found the volume of a box measuring $2\frac{1}{2}$ ft by 4 ft by 2 ft. He wrote: ' $V = 2 \times 4 \times 2 = 16$ cubic feet' and said the half 'was too small to matter.'

Explain Omar's misconception and give the correct volume.

Model answer: $V = \frac{5}{2} \times 4 \times 2 = 20$ cubic feet. The half foot is a real part of the edge; dropping it undercounts the volume by 4 cubic feet.

2 points	Student converts $2\frac{1}{2}$ to $\frac{5}{2}$, computes $\frac{5}{2} \times 4 \times 2 = 20$ cubic feet, and explains that dropping the half discards a real part of the length.
1 point	Student gives 20 cubic feet without explaining the conversion.
0 points	Student agrees with Omar or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9

WRITTEN RESPONSE · 2 POINTS

A toy company ships action figures in boxes that are $4 \times 3 \times 6$ inches. A shipping crate is $24 \times 12 \times 18$ inches. How many action figure boxes fit in the crate? Explain your reasoning step by step.

Model answer: Each toy box holds $4 \times 3 \times 6 = 72 \text{ in}^3$ and the crate holds $24 \times 12 \times 18 = 5184 \text{ in}^3$, so $5184 \div 72 = 72$ boxes fit. The division comes out exact because every crate edge is a whole-number multiple of a box edge: $24 \div 4 = 6$ across, $12 \div 3 = 4$ deep, and $18 \div 6 = 3$ layers, and $6 \times 4 \times 3 = 72$ confirms it.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.